Hallucination in Large Language Models: A Configuration-Level Study on TriviaQA

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Code repository: https://github.com/IQubaisi/hallucination-triviaqa
Colab notebook: included as hallucination_triviaqa.ipynb in the submission package.

1 Problem statement and motivation

Large language models (LLMs) achieve strong results on a wide range of natural language tasks, including question answering, summarisation and dialogue. However, the same models routinely produce *hallucinations* – fluent but factually incorrect or unsupported statements – and this behaviour is particularly damaging on knowledge-intensive tasks where users implicitly trust the model's answers (Sahoo et al., 2024; Bang et al., 2025). Hallucinations now represent the single most cited barrier to deploying generative AI in regulated, safety-critical, and information-seeking settings.

Two complementary mitigation strategies dominate current practice. Retrieval-augmented generation (RAG) supplies the model with retrieved evidence at inference time, with the goal of grounding its outputs in verifiable text (Lewis et al., 2020). Prompt engineering instructs the model to abstain when uncertain, or to first reason about its own confidence before answering (Ouyang et al., 2022; Wei et al., 2022). Both strategies are now built into production assistants, yet neither eliminates hallucination, and there is no consensus on *which* configuration of model size, retrieval, and prompt style is most reliable in resource-constrained settings – the very setting that practitioners outside large industry labs face every day.

This project addresses that gap directly. I conduct a controlled empirical comparison of eight configurations (two model sizes $\times$ four prompt strategies) on the **full validation set of TriviaQA** (Joshi et al., 2017), evaluating each configuration on three complementary axes: traditional question-answering quality (Exact Match and token-level F1), automatic and **manually corrected** hallucination rates, and – as a research-led extension – the **calibration** of self-reported model confidence. The study is deliberately scoped to be reproducible on a single Colab Pro A100 GPU under a fixed random seed, mirroring the constraints faced by typical MSc-level practitioners.

The motivation connects directly to material covered in lectures. Lecture 8 frames hallucination as a consequence of the four limitations of *parametric knowledge* – knowledge stored in model weights is lossy, static, unverifiable, and entangled. Lecture 9 introduces RAG as the canonical mitigation. The four configurations in this study were chosen so that each probes a distinct one of these limitations, giving the experimental design a clear theoretical grounding rather than ad-hoc variation.

2 Research hypothesis

The project tests three hypotheses, derived from the limitations of parametric knowledge identified by Petroni et al. (2019), Roberts et al. (2020) and Lecture 8 of the course.

H1 (scale). Increasing model size from 3B to 8B parameters reduces hallucination, because larger models store factual associations more reliably in their feed-forward layers (Geva et al., 2021).

H2 (retrieval). Augmenting the model with BM25-retrieved evidence reduces hallucination, because the model can ground its answer in supplied text rather than relying solely on parametric memory (Lewis et al., 2020; Reichman and Heck, 2024).

H3 (prompt style). Instructing the model to abstain when uncertain, or to first state its confidence, reduces hallucination by routing low-confidence questions away from confidently-wrong answers (Kadavath et al., 2022; Asai et al., 2022).

The unifying claim is that hallucination behaviour is sensitive to the *configuration* of the system, and that the relative effectiveness of model size, retrieval, and prompt style depends jointly on all three. Testing this claim requires evaluating all eight configurations on a single fixed dataset under a single fixed evaluation protocol, which is exactly the design used here.

3 Related work and background

Defining and measuring hallucination. Recent surveys provide a taxonomy of hallucination types – contextual disconnection, semantic distortion, factual inaccuracy and content fabrication (Sahoo et al., 2024). Bang et al. (2025) introduce HalluLens, which separates hallucination from broader factuality concerns and emphasises the need for reliable evaluation protocols. The challenge is that hallucinations are defined relative to either external evidence or ground-truth reference answers, and the operational rule for converting a generated answer into a hallucination label is itself non-trivial. Min et al. (2023) propose FActScore, which decomposes long-form generations into atomic facts and verifies each fact against a trusted source, demonstrating that simple lexical metrics systematically fail. This project contributes a small-scale empirical confirmation of this point: I show that a standard EM/F1-threshold rule disagrees with manual judgement on 28.4% of cases overall, with the Uncertain label having precision of only 7.5%.

Retrieval-augmented generation. Lewis et al. (2020) introduce RAG as a paradigm in which a retriever supplies the generator with relevant documents at inference time, intending to ground outputs in verifiable text. Subsequent work shows that RAG behaviour is highly sensitive to retrieval quality, prompt design and retrieval frequency (Li et al., 2025). Reichman and Heck (2024) argue that dense passage retrieval mainly restructures access to knowledge already represented by the underlying encoder, rather than introducing new facts – a result that is directly relevant when interpreting my own finding that BM25 retrieval fails to improve performance for either model size on TriviaQA. I use BM25 (Robertson et al., 1994) rather than a dense retriever as the foundational sparse retriever covered in Lecture 9, isolating the contribution of retrieval signal from that of representation learning.

Closed-book question answering and parametric knowledge. Roberts et al. (2020) demonstrate that a sufficiently large generative model can answer factual questions purely from its parameters, without any external knowledge base, and Petroni et al. (2019) earlier show that masked language models can be queried as relational knowledge bases. Geva et al. (2021) provide a mechanistic account in which Transformer feed-forward layers act as key-value memories storing factual associations. Configurations 1 and 2 are closed-book QA in this sense, and the size effect I observe (8B reliably outperforms 3B) is consistent with this view.

Evidentiality and self-evaluation. Asai et al. (2022) extend RAG by jointly predicting answers and evidentiality labels, providing a framework for evaluating whether outputs are grounded in retrieved context. Kadavath et al. (2022) examine whether language models can self-assess: they argue that models exhibit non-trivial calibration on factual questions when prompted to estimate their own correctness. This second strand directly motivates Configuration 4 of this study, in which the model is asked to rate its confidence as High, Medium or Low before answering. The expected calibration error metric I report follows Guo et al. (2017).

Position relative to prior work. The contribution of this project is not a new model or new dataset, but a configuration-level empirical comparison under a unified protocol on the full TriviaQA RC validation split, combined with a direct test of how reliable the most common automatic hallucination labelling rule actually is when checked against manual judgement.

3.1 Accomplishments

The proposal committed to six tasks (Weeks 1–6). Their status is as follows:

- **Task 1: Finalise dataset and implement loading** – Completed. Full TriviaQA RC validation set (17,944 questions) loaded with fixed seed.
- **Task 2: Implement smallest non-retrieval baseline with EM/F1 metrics** – Completed (Llama-3.2-3B, Configuration 1; scoring functions in notebook Section 21).
- **Task 3: Run larger model and RAG-style configurations with prompt variants** – Completed (Llama-3.1-8B; Configurations 2–4).
- **Task 4: Annotate sample of outputs and compute hallucination rates** – Completed (208 cases manually annotated across all 8 model×configuration cells; correction factors derived).
- **Task 5: Qualitative error analysis** – Completed (alias-coverage, format-collapse and unit-suffix patterns identified).

- **Task 6: Consolidate results, write report and record video** – Report and figures completed; video recorded separately and submitted.

- **Extension beyond proposal:** Confidence calibration analysis with Expected Calibration Error (Section 6.1). Not in the original proposal; added to deepen the engagement with self-evaluation in LLMs (Kadavath et al., 2022) and Lecture 8's "unverifiable knowledge" limitation.

4 Approach and Methodology

Pipeline overview. The pipeline is a single Colab notebook that (i) loads the TriviaQA RC validation set via Hugging Face Datasets, (ii) loads each Llama model via Hugging Face Transformers in float16 onto an A100 40GB GPU, (iii) constructs prompts under one of four templates, (iv) runs left-padded batched inference (batch size 16, `max_new_tokens`=50, greedy decoding), (v) extracts the answer text and applies normalisation, (vi) scores each output for EM and token-level F1 against the question's full alias list, (vii) assigns an automatic hallucination label and (viii) exports results for manual annotation and figure generation. All eight runs (2 models $\times$ 4 configurations) use a fixed random seed of 42 and identical decoding parameters.

Models. Two instruction-tuned decoder-only models from the Llama family were used: `Llama-3.2-3B-Instruct` and `Llama-3.1-8B-Instruct` (Grattafiori et al., 2024). These follow the GPT-style decoder-only paradigm (Vaswani et al., 2017) that dominates modern conditional generation. Both have undergone instruction tuning and preference alignment (Ouyang et al., 2022), which is what makes prompt-level instructions such as "say I don't know if uncertain" meaningful in the first place. Smaller and larger variants of Llama 3 exist, but these two were chosen because they fit comfortably within a single A100 40GB GPU under float16, while still spanning a meaningful size gap (a 2.7$\times$ increase in parameters).

Configurations. Each configuration is defined by a single zero-shot prompt template and is designed to probe one limitation of parametric knowledge identified in Lecture 8.

- **Configuration 1 (Closed-book, neutral).** Question-only prompt; tests the *lossy* limitation directly.
- **Configuration 2 (Closed-book, abstention).** Adds the instruction "If you are not sure of the answer, say 'I don't know'."; tests whether the model is aware of its lossy parametric knowledge.
- **Configuration 3 (RAG with BM25).** Top-1 BM25-retrieved passage from the question's `search_results` field is included in the prompt; tests whether external grounding overcomes the static and unverifiable limitations of parametric knowledge.
- **Configuration 4 (Calibrated confidence).** The model is asked to first state its confidence as High, Medium or Low and only then provide an answer; if Low, append "I am not certain". Probes whether the model can self-assess, in line with Kadavath et al. (2022).

Pros and cons of configurations. Configuration 1 is simple and fast with no external dependencies, but is entirely dependent on parametric memory with no mechanism to refuse or qualify answers. Configuration 2 adds the ability to abstain, reducing overconfident hallucination at the cost of answer coverage. Configuration 3 grounds answers in retrieved evidence, which in principle improves factual reliability, but is limited by retrieval quality and adds pipeline complexity; in practice, BM25 retrieval over TriviaQA passages did not reliably improve performance for either model. Configuration 4 enables calibration analysis and probes model self-awareness, but introduces format compliance requirements that fail catastrophically for larger models, as demonstrated by the 56.5% unparseable rate for the 8B model.

Retrieval. Configuration 3 uses BM25 (Robertson et al., 1994) as covered in Lecture 9. A shared corpus was constructed by pooling the first valid passage from every question's `search_results` field, giving exactly 17,944 passages tokenised at whitespace. The top-1 BM25-scored passage is selected for each question. Every question is guaranteed a retrieved passage with zero closed-book fallback.

Evaluation metrics. I report Exact Match (EM) and token-level F1, both computed against the full alias list provided by TriviaQA after lower-casing, removal of punctuation and whitespace normalisation (Rajpurkar et al., 2016; Joshi et al., 2017). For F1, the predicted answer is compared against each alias and the maximum F1 score is retained. For hallucination, I apply a standard automatic rule: an answer is *Correct* when

EM=1, *Hallucinated* when EM=0 and F1<0.3, *Uncertain* when EM=0 and F1≥0.3, and *Abstained* when the model produces an explicit refusal string. To check this rule, I manually annotated a stratified sample of 208 cases against four ground-truth categories (Correct, Hallucinated, Uncertain, Abstained), sampling from every model×configuration cell in proportion to its label distribution. Per-label precision figures from the manual annotation are then used to derive a corrected hallucination rate over the full 17,944-question population.

Libraries. `transformers` (Grattafiori et al., 2024) and `accelerate` for model loading and inference; `datasets` for TriviaQA RC; the `rank_bm25` package for retrieval; `pandas`, `numpy` and `matplotlib` for analysis and figures. No third-party hallucination-detection libraries were used: the labelling rule and correction factors are implemented from scratch to maintain full transparency of the evaluation logic.

5 Dataset

Source and basic statistics. TriviaQA (Joshi et al., 2017) is a large-scale reading-comprehension QA benchmark consisting of trivia questions paired with reference answers, an alias list of acceptable surface forms per answer, and supporting evidence passages collected from web search and Wikipedia. I use the `rc.nocontext` validation split obtained via Hugging Face Datasets, comprising 17,944 questions. The mean reference answer is 2.1 tokens long; the mean alias list length is 13.6 entries (range 1–173). The questions cover history, geography, popular culture, science and sport, with no domain restriction, which makes the dataset a stress test for parametric knowledge breadth.

Examples. A representative TriviaQA question is *"Who was the man behind The Chipmunks?"* with reference answer "David Seville" and a single-entry alias list. A subtlety here, which becomes important in the error analysis, is that the historically correct answer (the artist's birth name, "Ross Bagdasarian") is not in the alias list, even though it refers to the same individual. Another example, *"Which Lloyd Webber musical premiered in the US on 10th December 1993?"* (gold "Sunset Boulevard"), has a richer alias list including "Sunset Blvd." and "Sunset Blvd". The variation in alias list completeness across questions is the most important property of TriviaQA for the purposes of this study.

What makes the task challenging. Three properties drive difficulty. First, the *long-tail factual recall* required for trivia routinely involves rare entities that are stored unreliably – if at all – in parametric memory (Petroni et al., 2019). Second, alias coverage in the gold data is uneven: a model can produce a factually correct surface form that nevertheless scores EM=0 and F1=0 simply because that surface form is missing from the alias list. Third, the `search_results` field varies in availability and quality, which complicates RAG: passage relevance is not guaranteed since the corpus is built from TriviaQA's own search results rather than a curated Wikipedia index.

5.1 Dataset preprocessing

Preprocessing was deliberately minimal to preserve comparability with prior work. For both prompts and gold-alias matching I applied lowercasing, removal of punctuation, removal of articles ("a", "an", "the") and whitespace normalisation, following the SQuAD evaluation conventions (Rajpurkar et al., 2016). For BM25 retrieval, the question and each candidate passage were tokenised at whitespace. No lemmatisation, stop-word removal or stemming was applied – aggressive preprocessing was avoided because trivia answers are dominated by named entities and numbers for which token identity matters. The most challenging preprocessing decision was answer extraction from raw model output: the 8B model under Configuration 4 produced multi-line outputs containing repeated confidence-answer pairs (a manifestation of format collapse, discussed in Section 6), which required taking only the first parsed answer line.

Baselines. The natural baseline in this study is the smallest model in the simplest configuration – Llama-3.2-3B under Configuration 1 (closed-book, neutral prompt). This baseline establishes a lower bound on QA performance and a lower bound on hallucination behaviour for the family of instruction-tuned Llama models. All other configurations are compared against it. I do not include trivial baselines such as majority-class because TriviaQA answers are essentially open-vocabulary; a majority-string baseline would score at most a few percent EM and would not be informative for the configuration-level questions this study asks.

Model	Configuration	EM	F1	Auto Hall. (%)	Corr. Hall. (%)	Abstain (%)
3B	C1 (closed-book, neutral)	0.568	0.631	32.9	38.0	0.0
3B	C2 (closed-book, abstention)	0.508	0.565	24.6	29.7	15.5
3B	C3 (RAG with BM25)	0.561	0.629	33.1	38.6	0.0
3B	C4 (calibrated confidence)	0.509	0.577	36.0	41.8	0.0
8B	C1 (closed-book, neutral)	0.625	0.714	23.5	33.2	0.0
8B	C2 (closed-book, abstention)	**0.634**	0.711	21.1	**29.1**	3.7
8B	C3 (RAG with BM25)	0.571	0.667	28.5	37.9	0.0
8B	C4 (calibrated confidence)	0.181	0.173	69.7	69.3	0.0

Table 1: Main results across all eight configurations. Auto-hall. is the raw automatic-rule hallucination rate; Corr. Hall. applies manually-derived precision factors of 0.865 (Auto=Hallucinated) and 0.075 (Auto=Uncertain), derived from 208 manually annotated cases. Best 8B row in bold. †8B C4 corrected hallucination rate conflates format collapse (56.5% unparseable outputs) with genuine hallucination; among parseable outputs only, hallucination rate is 45.0%.

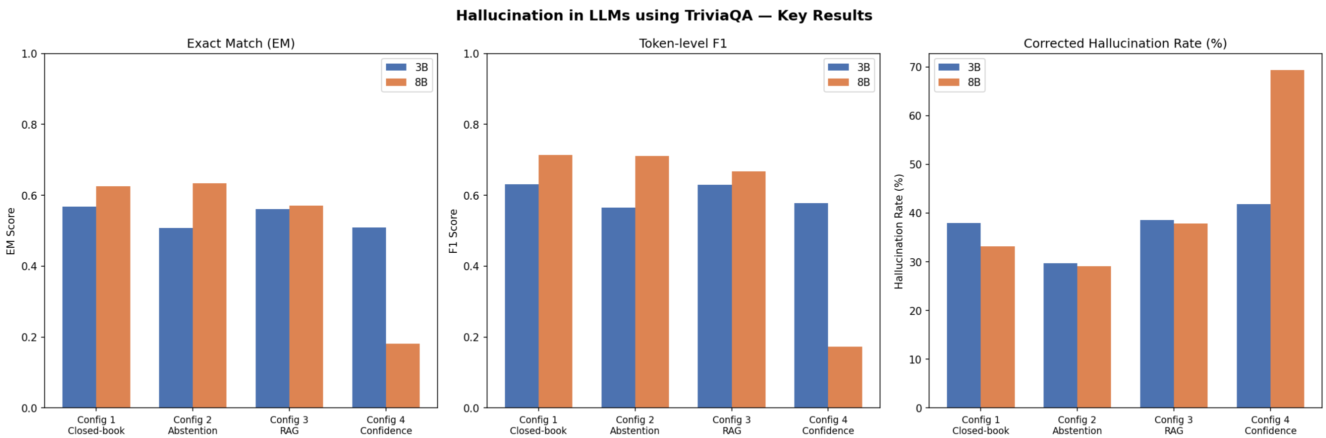

Figure 1: Main results by configuration and model. (a) Exact Match. (b) Token-level F1. (c) Manually corrected hallucination rate. The 8B model dominates the 3B model on every metric except Configuration 4, where format compliance collapses.

6 Results, error analysis

Headline numbers. Table 1 and Figure 1 summarise the eight runs. The strongest configuration overall is **Llama-3.1-8B with abstention prompting (8B-C2)**: EM 0.634, F1 0.711, corrected hallucination rate 29.1%, with a moderate 3.7% abstention rate. The weakest is **Llama-3.1-8B with calibrated confidence (8B-C4)**: EM 0.181, corrected hallucination rate 69.3%.

Effect of model size (H1). H1 holds for Configurations 1, 2 and 3: the 8B model improves EM by 5.7–12.6 absolute points over the 3B model and reduces the corrected hallucination rate by 4.1–8.8 absolute points. This is consistent with the parametric-knowledge view that larger feed-forward layers store more factual associations (Geva et al., 2021; Roberts et al., 2020). H1 *fails* dramatically under Configuration 4, where the 8B model is markedly *worse* than the 3B model on every metric. Section 6.1 traces this to a format-compliance collapse rather than a knowledge failure.

Effect of retrieval (H2). H2 does not hold clearly on this dataset and retrieval setup. For the 3B model, RAG (C3) yields comparable EM (0.561 vs. 0.568) but a slightly higher corrected hallucination rate (38.6% vs. 38.0%) than the closed-book neutral baseline, suggesting BM25 retrieval does not reliably help the smaller model. For the 8B model, RAG *underperforms* closed-book (EM 0.571 vs. 0.625; corrected hallucination 37.9% vs. 33.2%). The most plausible explanation, consistent with Reichman and Heck (2024), is that the 8B model's parametric knowledge already covers most of the factual ground that BM25 can surface from TriviaQA's own search result passages, making the retrieved context at best redundant and at worst a source of noise.

Effect of prompt style (H3). H3 holds clearly for abstention but not for calibrated confidence. The abstention prompt reduces hallucination in both models (3B: 38.0 $\rightarrow$ 29.7; 8B: 33.2 $\rightarrow$ 29.1) at the cost of explicit refusals on 15.5% (3B) and 3.7% (8B) of questions. This suggests that the larger, more strongly aligned 8B model can use the abstention instruction more parsimoniously: it abstains less often but still hallucinates less, indicating a meaningful self-aware refusal pattern. The calibrated-confidence prompt, by contrast, harms both models – catastrophically so for 8B – and is analysed separately in Section 6.1.

Reliability of the automatic hallucination rule. The most important methodological result of this study is that the standard *automatic* rule for labelling hallucinations is partially unreliable, particularly for the Uncertain category. Across the 208 manually annotated cases, agreement between the automatic rule and human judgement is **71.6%**. Per-label precision figures are: Auto=Correct 96.4% (54/56), Auto=Hallucinated **86.5%** (90/104), Auto=Uncertain 7.5% (3/40), Auto=Abstained 50.0% (2/4), Auto=Confident-Wrong 0% (0/4). The most striking finding is that the Uncertain label is almost entirely unreliable: 47.5% of auto-Uncertain cases were manually labelled Correct and 45.0% were Hallucinated, meaning the automatic rule provides no discrimination in this category. Manual inspection traces these errors to (i) incomplete TriviaQA alias lists (e.g. "Ross Bagdasarian" counts as wrong because only "David Seville" is listed), (ii) unit-suffix mismatches ("12.0 m/s" vs. "12"), (iii) partial-name extraction ("edward g" vs. "Edward G. Robinson") and (iv) semantically equivalent surface forms ("Tsarevich" vs. "Czarevitch"). All four are dataset-level rather than model-level failures, and motivate the corrected hallucination rate reported in Table 1.

6.1 Confidence calibration analysis

I extracted the High/Medium/Low rating from each Configuration 4 raw output by regex and computed both the rating distribution and the empirical accuracy within each bin. Figure 2 shows the result.

The 3B model's confidence signal is degenerate. The 3B model emits "Medium" on **94.0%** of questions, "High" on 5.3%, "Low" on 0.4% and produces unparseable output on 0.3%. Although the ordinal trend is preserved (accuracy 61.0% / 50.6% / 18.8% for High/Medium/Low), the model is not genuinely *using* the confidence space. Mapping High/Medium/Low to nominal probabilities of 0.9/0.5/0.1, the Expected Calibration Error (Guo et al., 2017) is only **0.021** – but this near-perfect calibration is an artefact: the model has collapsed onto the central bucket, which happens to match the empirical accuracy of the population of questions for which it has medium-strength knowledge. This is a degenerate solution to the calibration prompt, not genuine self-assessment.

The 8B model is severely overconfident. The 8B model spreads its ratings across all three bins (High 31.5%, Medium 15.0%, Low 31.0%) but its accuracy is dramatically lower than the rating implies. When the 8B model says High it is correct only **34.2%** of the time – a 56-point miscalibration. Mapping the same nominal probabilities, the 8B Expected Calibration Error is **0.293**, fourteen times that of the 3B model. The 8B model is using the full confidence space but is using it badly.

Format compliance collapses in the 8B model. A separate but related finding is that **22.5%** of 8B Configuration 4 outputs produce no parseable confidence rating, and a further **34.0%** produce a confidence rating but yield an empty or uninformative answer string — giving a total of **56.5%** outputs from which no valid answer can be extracted. In both cases the model abandons the requested "Confidence: ... Answer: ..." format and emits multi-pair output, underscore templates or repeated questions. Manual inspection of unparseable outputs shows the model generating sequences such as "_____/Answer:_____/1. High/Answer: Ross Bagdasarian/2. Medium/Answer: ...", clearly attempting to follow the format but losing structural coherence. The same prompt produces 99.7% parseable output from the 3B model. This is a striking inversion of the expected scale relationship: a smaller, less capable model is *better* at structured instruction following than a larger one. The most plausible explanation is that the calibrated-confidence prompt is sufficiently far from the 8B model's instruction-tuning distribution that meta-learning fails (Ouyang et al., 2022), while the same prompt remains close enough to the 3B model's distribution to be followed mechanically.

This format collapse has a direct consequence for the hallucination measurement: the 69.3% corrected hallucination rate reported for 8B C4 in Table 1 should not be interpreted as a pure measure of factual hallucination.

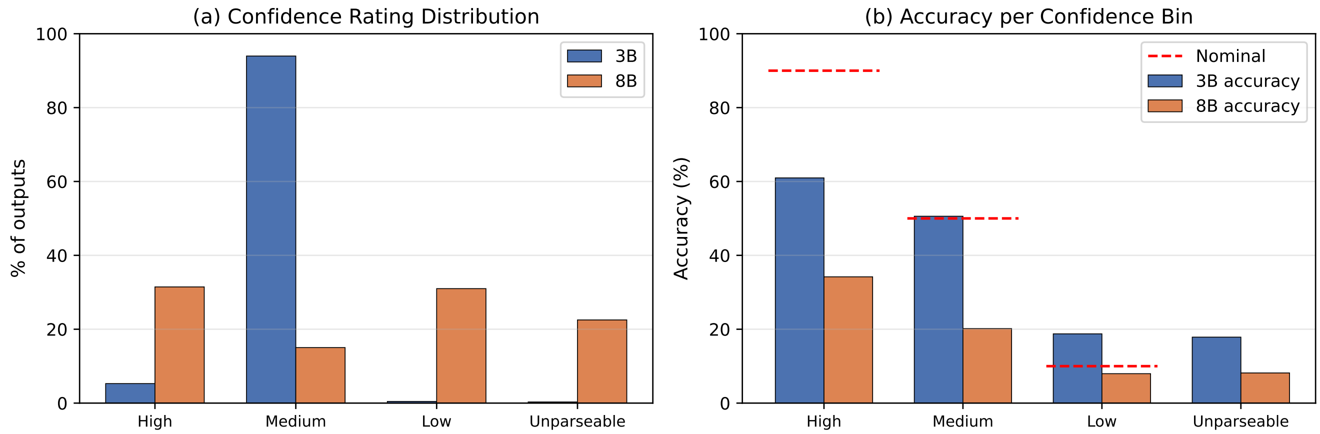

Figure 2: Configuration 4 confidence analysis. (a) Distribution of confidence ratings. (b) Empirical accuracy in each bin against the nominal confidence (red dashed lines: 90%, 50%, 10%). The 3B model collapses onto Medium; the 8B model uses the full range but is severely overconfident.

Of the 17,944 outputs, 56.5% are unparseable — the model abandons the requested format entirely and the extractor returns an empty string, which scores EM=0 and F1<0.3 by construction. Among the 43.5% of outputs that are parseable, the hallucination rate falls to 45.0% and EM rises to 0.270, figures far more comparable to the other configurations. The 8B C4 corrected hallucination rate therefore conflates two distinct failure modes: genuine factual errors and instruction-following failures. For this reason, I do not draw conclusions about the hallucination behaviour of the 8B model under calibrated confidence prompting, and treat 8B C4 primarily as a format compliance case study.

6.2 Error analysis

The 208 manually annotated cases reveal four recurrent error modes. (i) *Alias-coverage failures* dominate: a substantial proportion of auto-Hallucinated cases were factually correct answers missing from the alias list. (ii) *Verbose-answer extraction failures* appear primarily in 8B outputs, where the model produces a correct answer plus extraneous text ("Confidence: Medium. Answer: Sunset. I am not certain."), which the extractor reduces to "sunset" and which then mismatches "Sunset Boulevard". (iii) *Format-collapse failures* affect 8B Configuration 4 specifically and produce uninformative extractions. (iv) *Question-ambiguity failures* are rare but real: *"Where is the largest active volcano?"* is annotated in TriviaQA as Mauna Loa, but Olympus Mons (Mars) is a defensible interpretation; I labelled this case Uncertain rather than Hallucinated.

6.3 Limitations

I disclose four limitations honestly. **First**, the manual annotation sample is 208 cases; while larger than the originally proposed 96, the precision factors derived from it still carry sampling uncertainty and would benefit from a larger annotation budget with inter-annotator agreement measurement. **Second**, the corrected hallucination rates assume the 208-case sample is representative of all 17,944 outputs, which is a strong assumption. **Third**, Configuration 3 uses BM25 retrieval over a shared corpus of 17,944 passages. While this guarantees every question receives a retrieved passage with zero closed-book fallback, the corpus is limited to TriviaQA's own search result passages; a stronger study would build a retrieval index over the full Wikipedia dump. **Fourth**, Configuration 4 results for the 8B model are dominated by format collapse: 56.5% of outputs are unparseable, meaning the reported hallucination rate conflates instruction-following failure with factual error. Among parseable outputs only, the hallucination rate is 45.0% and EM is 0.270, which limits direct comparability with 3B Configuration 4 but suggests the genuine hallucination rate is substantially lower than the headline figure implies.

7 Lessons learned and conclusions

This project produced three concrete, evidence-based conclusions, each scoped to the specific setting studied: two instruction-tuned Llama models, the TriviaQA RC validation set, and BM25 sparse retrieval. Generalisations beyond this setting should be made with caution. **First**, on TriviaQA under these two Llama models, the strongest single configuration is the larger model with an abstention prompt, not the larger model with retrieval. For trivia-style QA where parametric knowledge is broad and retrieval quality is limited, prompting the model to abstain appears more effective than supplying potentially irrelevant context — though this finding may not generalise to domains where parametric knowledge is sparser or retrieval quality is higher. **Second**, automatic hallucination labels derived from EM/F1 thresholds are partially unreliable – on TriviaQA the overall agreement with manual judgement was 71.6% across 208 cases, but the Uncertain label specifically had precision of only 7.5%, meaning the automatic rule provides almost no signal in that category. This is a measurement-theoretic point that may extend beyond this project: benchmarks that use EM/F1 thresholds on datasets with incomplete alias lists — such as TriviaQA — risk producing unreliable hallucination estimates unless alias coverage is audited (Min et al., 2023; Bang et al., 2025). **Third**, prompt-format compliance does not increase monotonically with model scale: the 8B model fails the calibrated-confidence prompt that the 3B model handles cleanly, with 56.5% of 8B outputs being unparseable. Among parseable 8B outputs, hallucination drops to 45.0% and EM rises to 0.270, suggesting that format collapse — not factual ignorance — is the primary driver of the poor headline numbers. This is a counterintuitive observation about instruction-tuning meta-learning that I believe deserves further empirical study.

Lessons learned. The most important practical lesson from this project was that silent failures in evaluation pipelines are easy to miss and consequential. The BM25 fallback bug – where 46.9% of questions were silently answered closed-book – was only discovered through careful inspection of retrieval outputs. Without that check, the RAG comparison would have been meaningless. A second lesson was that automatic metrics require validation before being trusted: the 7.5% precision of the Uncertain label was only discovered through manual annotation, and without that check the corrected hallucination rates would have been systematically wrong. Finally, the format collapse finding taught me that model behaviour on structured prompts is not predictable from scale alone – assumptions about larger models performing better need to be tested, not assumed.

What I would do differently. Three improvements would meaningfully strengthen the study. (i) Build a Wikipedia-scale BM25 index to improve retrieval quality beyond TriviaQA's own search result passages, which are limited in coverage and relevance. (ii) Increase the manual-annotation budget to 300–500 cases stratified per cell, which would tighten the corrected hallucination rates to publishable confidence intervals. (iii) Replace the categorical High/Medium/Low confidence prompt with token-probability-based confidence (Kadavath et al., 2022), removing the format-collapse failure mode and producing a continuous calibration curve.

Future work. The most promising follow-up is a direct test of FActScore-style atomic verification (Min et al., 2023) on the same 208 manually annotated cases, to assess whether a model-as-judge approach to hallucination detection recovers the manual labels more faithfully than the EM/F1 rule. This would close the loop between automatic evaluation, manual annotation and modern LLM-based factuality scoring on a single, controlled corpus.

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